Exam

Name_____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) The term _____ can be used to describe anything that uses binary numbers. 1) _____
A) mnemonic B) processor C) computer D) digital

2) What is the largest value that can be stored in one byte? 2) _____
A) 255 B) 8 C) 128 D) 16

3) Each different type of CPU has its own _____. 3) _____
A) software B) machine language
C) syntax D) high-level language

4) The computer language that uses mnemonics for writing programs is _____. 4) _____
A) Java B) Assembly C) Ada D) Pascal

5) Which of the following is *not* an example of application software? 5) _____
A) Windows B) Microsoft Word C) PowerPoint D) a web browser

6) _____ refers to the physical components that a computer is made of. 6) _____
A) Software B) The system C) The device D) Hardware

7) _____ is the term used for a set of rules that must be strictly followed when writing a program. 7) _____
A) Key words B) Semantics C) Grammar D) Syntax

8) _____ is used to encode real numbers to store them in the computer's memory. 8) _____
A) Floating-point notation B) RAM
C) Assembly language D) Unicode

9) Computers can do many different jobs because they are _____. 9) _____
A) reliable B) automated C) electronic D) programmable

10) Which of the following is *not* an example of operating system software? 10) _____
A) Windows B) Microsoft Word C) Linux D) Mac OS X

11) A(n) _____ is a program that translates a high-level language into a machine language program. 11) _____
A) IDE B) CPU C) compiler D) storage device

12) Which of the following companies does *not* manufacture microprocessors? 12) _____
A) Dell B) Motorola C) Intel D) AMD

13) Which of the following functions does an interpreter perform with the instructions from a high-level language? 13) _____
A) It compiles and translates. B) It translates.
C) It translates and executes. D) It executes.
E) It compiles and executes. F) It compiles.

14) Software refers to _____. 14) _____
A) physical components of a computer B) programs
C) data stored in RAM D) the CPU

15) A computer program is _____. 15) _____
A) only used for desktop computers
B) the same as main memory
C) a set of instructions that allow the computer to solve a problem or perform a task
D) another name for an operating system

16) What is syntax? 16) _____
A) the words that have a special meaning in the programming language
B) the symbols or words that perform operations in a program
C) the words or characters that are defined by the programmer
D) the rules that must be followed when writing a program

17) The _____ cycle is used by the CPU to execute instructions in a program. 17) _____
A) decode - fetch - execute B) fetch - decode - execute
C) decode - execute - fetch D) execute - fetch - decode

18) A program is a sequence of instructions stored in _____. 18) _____
A) the computer's memory B) software
C) the CPU D) the operating system

19) The smallest storage location in a computer's memory is known as a _____. 19) _____
A) bit B) switch C) word D) byte

20) A(n) _____ is a software package that usually consists of a text editor, a compiler or interpreter, and tools to test and debug programs. 20) _____
A) CPU B) storage device C) compiler D) IDE

21) Programs that make a computer useful for everyday tasks are known as _____. 21) _____
A) IDEs B) operating systems
C) system software D) application software

22) USB drives store data in a special type of memory known as _____. 22) _____
A) flash memory B) disks
C) optical server memory D) magnetic memory

23) _____ is the coding scheme that contains a set of 128 numeric codes which are used to represent characters in a computer's memory. 23) _____
A) Two's complement B) Unicode
C) ASCII D) Binary numbering

24) The physical devices that make up a computer are its _____. 24) _____
A) hardware B) applications C) CPU D) software

25) The statements that a programmer writes in a high-level language are called _____. 25) _____
A) binary code B) source code C) processor code D) pseudocode

26) The program development cycle is made up of _____ steps that are repeated until no errors can be found in the program. 26) _____

A) four B) two C) five D) three

27) A(n) _____ is a diagram that graphically depicts the steps that take place in a program. 27) _____

A) pseudocode document B) flowchart
C) code list D) algorithm

28) A(n) _____ is a name that represents a value which cannot be changed during the program's execution. 28) _____

A) input variable B) uninitialized variable
C) named variable D) named constant

29) A(n) _____ symbol is used for an assignment statement in a flowchart. 29) _____

A) oval B) parallelogram C) processing D) triangle

30) Which of the following error types produces incorrect results but does not prevent the program from running? 30) _____

A) syntax B) human C) logic D) grammar

31) What is the error in the following pseudocode? 31) _____

```
Display "What is your name?"
Input userName
Declare String userName
```

A) `userName` has been used before it is declared.
B) The `Input` statement should be the first statement.
C) `userName` is an invalid variable name.
D) There is no error.

32) _____ is the informal language used by programmers to create models of programs. 32) _____

A) Pseudocode B) Algorithm C) Source code D) Flowcharting

33) What term is used for a string that appears in the actual code of a program? 33) _____

A) string literal B) str C) virtual string D) any of these

34) What is the value of the variable `result` in the following expression? 34) _____

```
Set result = (6 + 8) / 4 * 2
```

A) `22` B) `10` C) `7` D) `28`

35) What is the value of the variable `result` in the following expression? 35) _____

```
Set result = 6 + 8 * 4 / 2
```

A) `7` B) `28` C) `22` D) `10`

36) The _____ operator is used to raise 5 to the second power. 36) _____

A) `*` B) `MOD` C) `/` D) `^`

37) What is the error in the following pseudocode? 37) _____

```
Declare Integer widgets
Declare Real cost
Set widgets = 3.5
Set cost = widgets * 5
```

A) `widgets` is an invalid variable name for this code.
B) `widgets` has been declared as an `Integer` and cannot hold a floating-point value.
C) You cannot mix a `Real` variable and an `Integer` variable in a program.
D) There is no error.

38) _____ is the encoding technique used to store negative numbers in the computer's memory. 38) _____
A) ASCII
B) Two's complement
C) Floating-point notation
D) Unicode

39) A computer system consists of all of the following *except* _____. 39) _____
A) main memory
B) the operating system
C) input and output devices
D) the CPU

40) The first high-level programming language that could perform complex mathematical calculations was _____. 40) _____
A) C++ B) COBOL C) FORTRAN D) Ada

41) Main memory is also called _____. 41) _____
A) RAM
B) secondary storage
C) the hard drive
D) ROM

42) The following is an example of an instruction written in the _____ computer language. 42) _____

```
1011000 010100110
```

A) Assembly
B) machine language
C) C++
D) Java

43) Application software refers to _____. 43) _____
A) mobile devices
B) key words
C) the operating system
D) programs that make the computer useful to a user

44) The process of stepping through each of a program's statements, one by one, to see what each statement does is known as _____. 44) _____
A) bug checking B) executing C) debugging D) hand tracing

45) The _____ structure consists of a set of statements that execute in the order in which they appear. 45) _____
A) sequence B) decision C) repetition D) control

46) A variable declaration typically specifies the variable's _____ and _____. 46) _____

A) number, name
B) name, value
C) data type, value
D) name, data type

47) _____ is an extensive encoding scheme that is compatible with ASCII. 47) _____

A) Assembly language
B) Floating-point notation
C) RAM
D) Unicode

48) The following is an example of a(n) _____ statement. 48) _____

```
Set rate = 6.25
```

A) assignment B) output C) declaration D) input

49) What is the value of the following expression? 49) _____

```
12 - 4 * 3 / 2 + 9
```

A) `21` B) `-6` C) `2.18` D) `15`

50) What is the error in the following pseudocode? 50) _____

```
Declare String user
Display "How many widgets do you want to buy?"
Input user
```

A) `user` is an invalid variable name for this code.
B) `user` has been used before it is declared.
C) The input is a number, not a string.
D) The `Input` statement should be the first statement.

Answer Key
Testname: ONESYS SAMPLE

1) D
2) A
3) B
4) B
5) A
6) D
7) D
8) A
9) D
10) B
11) C
12) A
13) C
14) B
15) C
16) D
17) B
18) A
19) A
20) D
21) D
22) A
23) C
24) A
25) B
26) C
27) B
28) D
29) C
30) C
31) A
32) A
33) A
34) C
35) C
36) D
37) B
38) B
39) B
40) C
41) A
42) B
43) D
44) D
45) A
46) D
47) D
48) A
49) D
50) C